Joint Impact of Fog and Aerosol Particle Size Distribution In LiDAR-Based 3D Object Detection Systems

Prithvi Singh Dangas, *Student ABV-IIITM Gwalior*, Gaurav Sharma, *Member, IEEE*, Manisha Pattanaik, *Senior Member, IEEE*, Sri Niwas Singh, *Fellow, IEEE*

***Abstract*—Recent studies have highlighted the vulnerability of LiDAR-based 3D object detection systems to adverse weather conditions, particularly fog and smog. While prior work has introduced heuristic degradation models or high-level sensor fusion strategies, few have explored the fine-grained physical mechanisms by which fog particles attenuate and scatter LiDAR signals. This project addresses this gap by modeling the impact of fog using physically grounded scattering theory, specifically leveraging the joint particle size distribution of fog and smog to calculate an accurate attenuation constant (β). These models are used to synthesize realistic foggy LiDAR point clouds from clear-weather datasets such as KITTI.The proposed simulation framework integrates particle-level Mie scattering effects into a fog injection pipeline, enabling high-fidelity degradation of point cloud data. We evaluate the downstream impact on 3D object detection using PV-RCNN and other state-of-the-art models, quantifying performance drops and highlighting failure modes under varying fog densities. Our results reveal the limitations of current perception algorithms when exposed to physically realistic weather-induced distortions and demonstrate the value of physically simulated data for training and evaluation. By anchoring fog simulation in validated atmospheric science, this work provides a more principled foundation for adverse weather perception and lays the groundwork for future integration with sensor fusion, SLAM, and dynamic environment modeling.**

***Index Terms*—3D Object Detection, LiDAR, Fog Simulation, Mie Scattering, Autonomous Driving, Adverse Weather, PV-RCNN, Particle Size Distribution.**

I. Introduction

Adverse weather conditions, particularly those involving particulate matter such as fog and aerosols, present significant challenges to the reliable operation of autonomous vehicles. The degradation of sensor measurements under such environmental influences directly impacts perception systems, which are critical for safe navigation and decision-making. Among the various sensing modalities, Light Detection and Ranging (LiDAR) systems have emerged as a cornerstone technology for three-dimensional (3D) environment understanding due to their ability to provide precise geometric information. However, LiDAR performance is inherently sensitive to atmospheric phenomena, where the scattering and absorption of laser pulses by suspended particles can distort the captured point clouds.

Traditional approaches to modeling weather-induced sensor degradation often rely on simplified assumptions, such as uniform noise injection or isotropic attenuation models, that fail to capture the complex physical interactions occurring between laser beams and atmospheric constituents. In particular, the scattering behavior is highly dependent on the size distribution, concentration, and composition of the particles suspended in the medium. Fog particles, typically larger and more homogenous, and aerosols, often smaller and more heterogeneous, exhibit distinct scattering characteristics governed by Mie theory. Their combined presence in smog conditions leads to a multifaceted degradation profile that cannot be accurately represented by simplistic models.

Despite the growing body of research on robustness of deep learning-based 3D object detectors to adverse weather, relatively little attention has been devoted to physically grounded modeling of environmental effects at the LiDAR measurement level. Existing datasets and simulation frameworks inadequately address the coupled influence of fog and aerosol particles on LiDAR signals, limiting the understanding of how specific environmental parameters affect detection performance.

In this paper, we present a comprehensive framework to model and analyze the joint impact of fog and aerosol particle size distributions on LiDAR-based 3D object detection systems. By employing a physically inspired model rooted in Mie scattering theory, we construct a joint probability distribution that characterizes the coexistence of fog and aerosol populations under smog conditions. This distribution is utilized to synthesize realistic degradation effects on LiDAR returns, thereby enabling a more faithful simulation of adverse environmental conditions.

Our methodology is applied to augment the KITTI dataset, a widely used benchmark in autonomous driving research, with controlled smog-induced distortions. This augmentation enables systematic evaluation of 3D object detectors under varying atmospheric conditions, providing novel insights into the resilience and failure modes of perception algorithms. To the best of our knowledge, this work represents the first effort to incorporate a joint fog-aerosol size distribution into the modeling of LiDAR degradations for 3D object detection tasks.

The main contributions of this paper are as follows:

- We develop a joint modeling framework that integrates fog and aerosol particle size distributions to simulate smog-induced LiDAR signal distortions based on fundamental scattering physics.
- We introduce a new dataset augmentation pipeline for the KITTI dataset, enabling rigorous evaluation of 3D object

detectors under physically realistic smog conditions
- We analyze the relationship between particle size characteristics and the degradation of 3D object detection performance, providing insights into the physical factors that impact autonomous vehicle perception in adverse weather.

By bridging atmospheric physics with 3D computer vision, this work advances the understanding of environmental robustness in autonomous systems and lays the groundwork for future research in physically informed sensor modeling and resilient machine perception.

II. Related Work

Recent advancements in 3D object detection using LiDAR sensing have increasingly focused on robustness under adverse weather conditions. For instance, Zhang et al. [1] proposed an uncertainty-aware 3D object detection framework leveraging sensor fusion to mitigate degradation due to environmental disturbances. Similarly, Chen et al. [2] introduced a LiDAR denoising and restoration model specifically designed for fog and snow scenarios, significantly enhancing detection performance in corrupted point clouds. In a complementary effort, Wang et al. [3] developed physics-inspired augmentation techniques to simulate the impact of scattering and absorption phenomena on LiDAR returns, providing more realistic training data.

Studies such as Li et al. [4] emphasized adaptive point cloud processing under dynamic weather conditions by integrating spatiotemporal consistency modeling into LiDAR perception pipelines. Furthermore, Yang et al. [5] explored multi-task learning approaches combining detection and weather estimation tasks to enable environment-aware object detection systems. These recent contributions underscore the growing recognition of atmospheric challenges in autonomous driving and highlight the need for physically-grounded models that faithfully capture adverse environmental effects.

LiDAR technology itself has also witnessed rapid advancements, both in hardware and software domains, facilitating higher-resolution, denser, and more reliable sensing capabilities critical for autonomous navigation. Recent works such as Sun et al. [6] introduced RSN (Range Sparse Net) to better handle sparse point clouds by learning structured range representations. Meanwhile, Bai et al. [7] proposed PointAugmenting to enrich point cloud features by generating additional semantically coherent points, significantly boosting downstream detection accuracy. Advances in self-supervised learning for LiDAR, such as OccFlow [8], have also enabled enhanced scene understanding by predicting dynamic occupancy flow from raw point clouds. These studies collectively emphasize not only the importance of robust 3D detection under challenging conditions but also the continual push towards maximizing the information extracted from LiDAR returns, even under sparsity or corruption.

The impact of environmental disturbances on LiDAR sensing has gained considerable attention in recent literature. Studies such as those by Bijelic et al. [9] and Sakaridis et al. [10] examined the effects of rain, fog, and snow on perception systems, often modeling such conditions as sources of noise or occlusion. Complementary work by Heinzler et al. [11] provided a comprehensive analysis of weather-induced point cloud distortions through data-driven simulations, while Kilic et al. [12] developed a physically based attenuation model specifically for automotive LiDAR systems under fog conditions. Notably, Hahner et al. [13] advanced fog simulation techniques by synthesizing realistic fog effects on real LiDAR point clouds, enabling targeted evaluation of 3D object detection robustness in adverse weather. Similarly, Peng et al. [14] investigated the performance of parallel LiDAR configurations in foggy conditions, offering insights into sensor fusion strategies to mitigate weather-related degradation. While these efforts provide valuable empirical insights, they frequently abstract environmental effects using coarse stochastic models or focus on system-level adaptations, often without explicitly considering the fine-grained physical scattering mechanisms associated with specific particle size distributions and their joint impact on detection performance.

In the broader field of optical sensing, the theoretical underpinnings of light scattering by particles have been well established through Mie theory [15], [16]. These models, extensively validated in atmospheric science, account for the size, refractive index, and morphology of particles relative to the incident wavelength. While applications of Mie scattering have been explored in LiDAR remote sensing for atmospheric monitoring [17], their integration into autonomous vehicle perception pipelines remains relatively nascent. Venugopal et al. [18] recently introduced light scattering models tailored for automotive LiDAR systems; however, their work primarily emphasized scattering intensity characteristics rather than downstream object detection performance.

Parallel efforts in synthetic data generation, such as those reported in [19], [20], have incorporated simulated adverse weather conditions to enhance the robustness of perception models. Although these domain adaptation strategies demonstrate notable improvements, they often rely on heuristic approximations of weather effects and neglect the nuanced influence of particle size distributions on LiDAR point returns.

More physically grounded modeling approaches, such as those leveraging radiative transfer simulations [21], have begun to appear, yet they are predominantly geared towards sensor modeling fidelity rather than end-to-end evaluation within object detection frameworks. As a result, the detailed characterization of how the joint distribution of fog and aerosol particle sizes influences 3D object detection accuracy remains an underexplored area.

The present work seeks to bridge this gap by systematically modeling the effects of smog—comprising both fog and aerosol constituents—on LiDAR-based 3D object detection. By integrating physically-validated particle scattering models with real-world datasets, we provide a comprehensive analysis of the degradation mechanisms at play, thereby offering new insights into the design of more resilient perception systems for autonomous driving applications.

III. Methodology

A. LiDAR in Clear Weather

In clear weather conditions, LiDAR performs optimally, providing high-resolution and accurate data, which is crucial for various applications, including autonomous driving, environmental monitoring, and urban planning.

The received power in LiDAR Sensor is mathematically represented as:

$$P_r(R) = C \int_0^{\frac{2R}{c}} P_t(t) H\left(R - \frac{ct}{2}\right) dt, \tag{1}$$

where $P_r(R)$ is the received power as a function of range R, $P_t(t)$ is the time-dependent transmitted signal power, c is the speed of light, C is the sensor constant independent of time and range, and $H(.)$ represents the total impulse response of the medium.

$P_t(t)$ is mathematically expressed as:

$$P_t(t) = \begin{cases} P_0 \sin^2\left(\frac{\pi t}{2\tau_h}\right), & \text{if } 0 \le t \le 2\tau_h, \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

where P_0 represents the peak power of the transmitted pulse, and τ_H denotes the half-power pulse width. The overall impulse response $H(R)$ of the LiDAR-based system is mathematically formulated as:

$$H(R) = H_o(R) H_t(R), \tag{3}$$

where $H_t(R)$ is the distance-based channel impulse response of the target, and $H_o(R)$ is the impulse response of the optical channel, expressed as:

$$H_o(R) = \frac{T^2(R)}{R^2} \xi(R). \tag{4}$$

From Equation (4), $T(R)$ is the total one-way transmission loss, and $\xi(R)$ represents the crossover function which defins the ratio of the area illuminated by the transmitter to the area observed by the receiver of the LiDAR sensor. $T(R)$ is represented by:

$$T(R) = \exp\left(-\int_0^R \alpha(r) dr\right), \tag{5}$$

α is the attenuation coefficient. For homogeneous conditions, this simplifies to:

$$T(R) = \exp(-\alpha R). \tag{6}$$

For clear weather, the value of H_t is given as:

$$H_t(R) = H_t^{\text{Hard}}(R) = \beta_0 \delta(R - R_0), \tag{7}$$

β_0 is the differential reflectivity of the target, and $\delta(.)$ is the Kronecker delta function.In clear weather condition, H_C is represented as:

$$H_c(R) = \frac{\xi(R)}{R^2}. \tag{8}$$

By substituting Equations (7) and (8) into (1), the received power in clear weather condition is mathematically given as:

$$P_{r,cl}(R) = C_A \int_0^{2\tau_H} P_0 \sin^2\left(\frac{\pi t}{2\tau_H}\right) \times \beta_0 \delta\left(R - R_0 - \frac{ct}{2}\right) dt. \tag{9}$$

Applying algebraic manipulation, the closed-form expression for received power is:

$$P_{R,CL}(R) = \begin{cases} \frac{C_A P_0 \beta_0}{R_0^2} \\ \times \sin^2\left(\frac{\pi(R-R_0)}{c\tau_H}\right), & \text{if } R_0 \le R \le R_0 + c\tau_H, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

B. LiDAR in Adverse Conditions

LiDAR sensor data is significantly impacted by attenuation and backscattering in adverse weather conditions.Since a precise model for calculating attenuation and backscattering coefficient is unavailable in the existing literature,so to characterize these phenomena Mie theory is used.

To realistically simulate the effects of fog in the channel, the channel impulse response is divided into contributions from soft and hard targets.

$$H_t(R) = H_t^{\text{Soft}}(R) + H_t^{\text{Hard}}(R), \tag{11}$$

where the soft target response for fog particles is given as:

$$H_t^{\text{Soft}}(R) = \beta U(R - R_0). \tag{12}$$

By combining the equations, the total impulse response becomes:

$$H(R) = \frac{\exp(-2\alpha R)\xi(R)}{R^2} \left(\beta U(R - R_0) + \beta_0 \delta(R - R_0)\right). \tag{13}$$

The total received power, accounting for both hard and soft targets, is represented as:

$$P_{r,\text{fog}}^{\text{Hard}}(R) = C P_0 \beta \frac{\exp(-2\alpha R_0)}{R_0^2} \int_0^{2\pi i} P_0 \sin^2\left(\frac{\pi t}{2\pi\mu}\right) \times \beta_0\, \delta\left(R - R_0 - \frac{ct}{2}\right) dt, \tag{14}$$

$$= \exp(-2\alpha R) P_{r,cl}(R).$$

For the particulate matter (combination of fog and aerosol), the received power is modeled as:

$$P_{r,\text{fog}}^{\text{Soft}}(R) = C \frac{\exp(-2\alpha R_0)}{R_0^2} \int_0^{2T_H} P_0 \sin^2\left(\frac{\pi t}{2T_h}\right) \times \xi(R - \frac{t}{2}) \beta_0\, \delta\left(R_0 + \frac{ct}{2} - R\right) dt, \tag{15}$$

The total received power at the LiDAR receiver is obtained by combining the hard and soft target responses:

$$P_{r,\text{fog}}(R) = P_{r,\text{fog}}^{\text{Hard}}(R) + P_{r,\text{fog}}^{\text{Soft}}(R). \tag{16}$$

Approximately the value of the backscattering coefficient (β) in terms of MOR is given as:

$$\beta = \frac{0.046}{MOR}. \tag{17}$$

C. Effect of Joint Distribution For Smog(Fog+Aerosol) Particles

The scattering of laser beams from the surface of fog and aerosol particles can be modeled by assuming the droplets are liquid spheres. Based on this assumption, the extinction efficiency, Q_{ext}, and the backscattering efficiency, Q_b, both functions of the diameter of fog droplets and aerosol, can be calculated using Mie Theory. Throughout the manuscript, we consider an optical transmitting source with a wavelength of 905 nanometers (nm). The values of both the attenuation and backscattering coefficients are shown in Fig. 1.

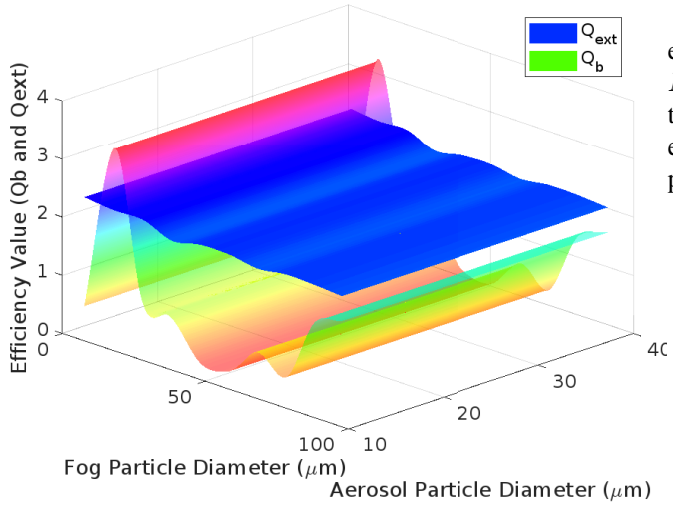


Fig. 1: Efficiency values of Q_{ext} and Q_b as a function of fog and aerosol diameter.

Fig. 1 shows a 3D surface plot illustrating the scattering efficiency Q_b and Q_{ext} as a function of two parameters,x_1 and x_2 Where x_1 is the diameter of fog particles and x_2 is the diameter of aerosol particle. The color map represents the magnitude of the scattering efficiency Q_b, with the highest efficiencies shown in red and the lowest in blue, whereas the red surface map represents the magnitude of extinction efficiency Q_{ext}.

The generalised gamma (GG) joint particle size distribution is used to model particle size variations for fog and aerosol environments. The distribution parameters were determined based on the given radius, shape, and scaling parameters.

The joint probability density function (pdf) of two correlated GG distributed random variables, representing the diameter of fog and aerosol particles, is given by:

$$N(x_1,x_2) = \frac{4\beta^2 m^{m+1}(x_1x_2)^{\beta(m-1)}(1-\rho)^{(1-m)/2}}{(\alpha_1\alpha_2)^{(m+1)/2}(1-\rho)\Gamma(m)} \times \exp\left(-\frac{m}{1-\rho}\left(\frac{x_1^2}{\alpha_1}+\frac{x_2^2}{\alpha_2}\right)\right) \times I_{m-1}\left(\frac{2m(x_1x_2)^{\beta}}{1-\rho}\sqrt{\frac{\rho}{\alpha_1\alpha_2}}\right), \quad (18)$$

where m and $m \geq 1/2$ are the distribution shaping parameters.$\alpha_\ell = \mathbb{E}(X_\ell^2) > 0$ is the scaling parameter with $\mathbb{E}(\cdot)$ denoting expectation.$I_{m-1}(\cdot)$ is the $(m-1)$-th order modified Bessel function of the first kind.$\Gamma(\cdot)$ is the Gamma function.The correlation coefficient between X_1^2 and X_2^2 is related to the correlation coefficient between Nakagami-m rvs.

TABLE I: Fog and Aerosol Gamma Distribution Parameters

Parameter	Value	Description
x_1	1×10^{-6} to 100×10^{-6} m	Fog diameters
x_2	12×10^{-6} to 20×10^{-6} m	Aerosol diameters [22]
m	3	Shape parameter
β	1	Gamma parameter
α_1	3.367×10^{-9} m^3	Scaling parameter fog
α_2	2.613×10^{-10} m^3	Scaling parameter aerosol
ρ	0.5	Correlation coefficient

By substituting the parameter values from Table I into eq. (18), the fog and aerosol particle size distribution, $N(D)(x1,x2)$ is calculated using the joint distribution of two correlated Gamma distributed random variables. Mathematically, the joint distribution of correlated fog and aerosol particles is expressed as:

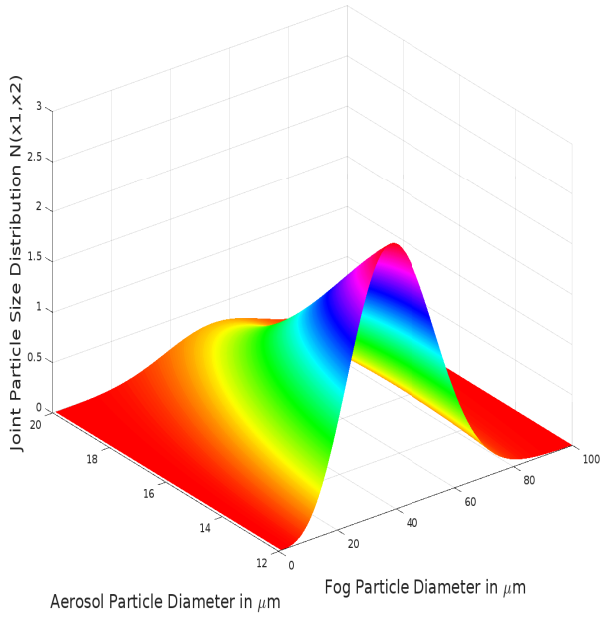


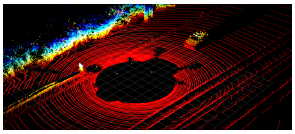
Fig. 2: Fog particle size distribution using GG Joint distribution.

Fig. 2 displays a 3D surface plot of $N(x_1,x_2)$ which represents a normalized density function, plotted against x_1 and x_2.Where x_1 is the diameter of fog particles and x_2 is the diameter of aerosol particles.

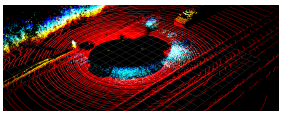
Assuming non-elastic scattering, the attenuation (α) and the backscattering (β) coefficients in terms of the smog(fog+aerosol) particle size distribution are mathematically represented as

$$\alpha = \frac{\pi}{8}\int_{D=0}^{\infty} D^2 Q_{\text{ext}}(D)N(D)\,dD, \quad (19)$$

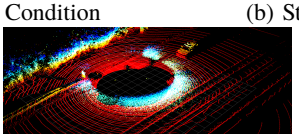
$$\beta = \frac{\pi}{8}\int_{D=0}^{\infty} D^2 Q_b(D)N(D)\,dD, \quad (20)$$



(a) Clear Weather Condition



(b) Strong MOR Fog



(c) Smog(Fog+Aerosol)

Fig. 3: Point clouds in different foggy environments: (3.a.) Shows LiDAR sensor visualization in Clear weather condition β (3.b.) Shows strong advection fog with β from (16); (3.c.) Shows smog(fog+aerosol) using β from (18).

where $N(D)$ is the probability of hitting a particle of diameter D.

TABLE II: Values of α and β in Strong and Moderate Advection Fog.

Weather Conditions	α	β	β **calculation**
Strong Fog	0.028995	0.0011	Using MOR.
Smog(Fog+Aerosol)	0.04009	0.03834	Using Joint Distribution.

For a detailed qualitative analysis, the point cloud scene representation of the LiDAR sensor data is shown in Fig. 3.In the first image (a), captured under clear weather, the LiDAR returns are strong and well-defined, allowing for precise scene mapping without interference. The second image (b) shows the impact of strong meteorological optical range (MOR) fog, where dense fog particles scatter and attenuate the LiDAR signals, resulting in reduced point density and clarity. The third image (c) depicts a smoggy environment, combining the effects of both fog and aerosol particles, which further degrade the LiDAR signal by enhancing scattering and absorption, modeled using an advanced backscattering coefficient. Collectively, these images highlight the challenges LiDAR sensors face in adverse weather, critically important for autonomous driving systems.

IV. Results and Discussions

Table III presents a comparative evaluation of the mean Average Precision (mAP) scores achieved by several state-of-the-art detection models on the KITTI dataset under varying environmental conditions, specifically clear weather, fog, and smog. The evaluation is conducted across three object categories—Car, Cyclist, and Pedestrian—and three levels of difficulty: easy, moderate, and hard.

Notably, YOLOv7 and YOLOv11 are included in the evaluation to provide a comparative baseline for 2D object detection models, contrasting their performance with dedicated 3D object detection models such as VOXEL-RCNN, PV-RCNN, and SECOND. Under clear weather conditions, YOLOv7 and YOLOv11 achieve exceptionally high Car AP scores of 98.6% and 98.4%, respectively, on the easy difficulty subset. However, their performance on the Cyclist and Pedestrian categories, while relatively strong, still indicates challenges in detecting smaller or less distinct objects, particularly under adverse conditions.

The 3D detection models demonstrate competitive performance in clear conditions, with PV-RCNN achieving 94.3% Car AP on moderate difficulty and 93.3% on hard difficulty, and VOXEL-RCNN reaching 89.3% on moderate. However, all models experience notable degradation under fog conditions. For instance, PV-RCNN's Car AP decreases to 89.9% on moderate and 87.8% on hard subsets under fog, while Cyclist and Pedestrian AP scores drop significantly—highlighting the increased difficulty of perception in particulate weather.

The impact of smog, which introduces combined fog and aerosol scattering effects, is even more severe. PV-RCNN's Car AP under smog conditions falls to 88.2% (moderate) and 86.8% (hard), while Cyclist and Pedestrian APs are further diminished, reaching 35.2% and 20.05%, respectively. SECOND similarly shows substantial performance drops under smog, with the Pedestrian AP declining to 17.94% (moderate).

Overall, the results underscore two important observations:

- While 2D detectors such as YOLOv7 and YOLOv11 achieve strong baseline performance in clear conditions, 3D detectors offer structural advantages for spatial reasoning under normal circumstances.
- Adverse environmental conditions—particularly smog—cause severe degradation across both 2D and 3D models, emphasizing the critical need for robust, physically-informed perception strategies in autonomous driving.

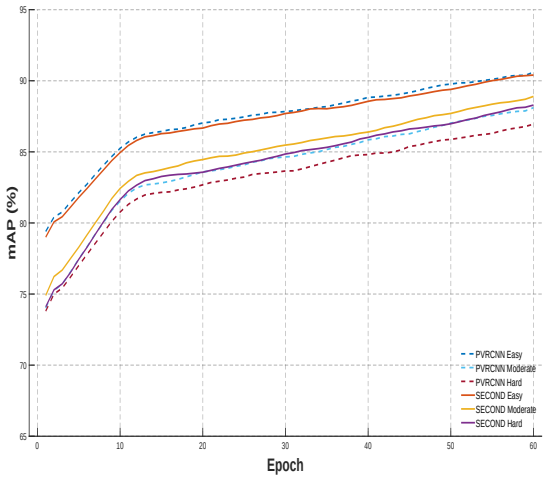


Fig. 4: Car Class mAP vs Epochs for PVRCNN and SECOND

Fig. 4 presents the mean Average Precision (mAP) values for the Car class under easy, moderate, and hard detection difficulty levels using PVRCNN and SECOND architectures. As depicted, the performance of the system improves with easier detection conditions, achieving the highest mAP values under the easy setting. Specifically, PVRCNN attains mAP values of 90.51%, 88.2%, and 86.95% for easy, moderate, and hard conditions, respectively, while SECOND achieves slightly higher mAP values of 90.43%, 88.9%, and 88.29%. The superior performance under easy conditions can be attributed to clearer object visibility and minimal occlusion, resulting in

TABLE III: Mean Average Precision (mAP) of different methods on KITTI in different weather conditions

Method	Car AP			Cyclist AP			Pedestrian AP		
	easy	moderate	hard	easy	moderate	hard	easy	moderate	hard
YOLOv7 [23] (Clear Weather)	98.6	-	-	93.2	-	-	86.4	-	-
YOLOv11 [24] (Clear Weather)	98.4	-	-	87.9	-	-	88.1	-	-
VOXEL-RCNN (Clear Weather)	97.2	89.3	89.1	-	-	-	-	-	-
PV-RCNN (Clear Weather)	98.2	94.3	93	83.7	71.8	67.6	63.4	57.08	63.1
PV-RCNN (Fog Condition)	90.89	89.9	87.8	41.2	37	33.2	56.88	47.5	43.4
PV-RCNN (Smog Condition)	90.51	88.2	86.95	40.04	34.93	33.56	19.76	15.86	14.5
SECOND (Smog Condition)	90.43	88.9	88.29	41.9	35.7	33.9	21.35	17.94	16.48

richer point cloud representations. In contrast, moderate and hard conditions introduce greater environmental complexity and occlusion, reducing the number of points representing the car and subsequently leading to a slight decrease in detection performance. Nevertheless, both PVRCNN and SECOND demonstrate robust and consistent performance across all difficulty levels, maintaining high mAP values even in challenging detection scenarios.

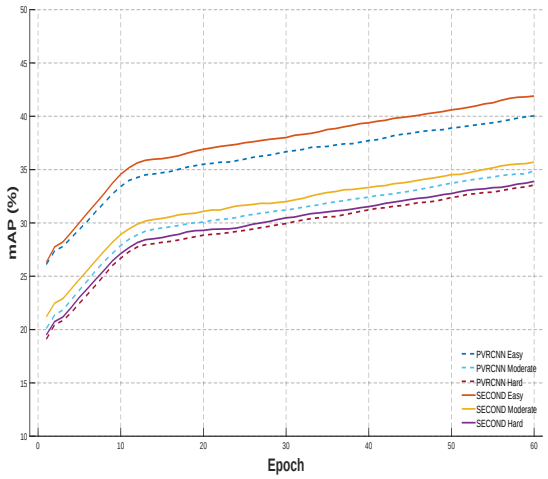


Fig. 5: Cyclist Class mAP vs Epochs for PVRCNN and SECOND

Fig. 5 further illustrates the mean Average Precision (mAP) values for the Cyclist class under easy, moderate, and hard detection difficulty levels using PVRCNN and SECOND architectures. From the plots, it is evident that detection performance decreases as the difficulty increases. Specifically, PVRCNN achieves mAP values of 40.04%, 34.9%, and 33.56% for easy, moderate, and hard settings, respectively. Meanwhile, SECOND outperforms PVRCNN with mAP values of 41.9%, 35.7%, and 33.9% for the corresponding difficulty levels. The overall lower mAP values compared to the Car class can be attributed to the smaller size and higher mobility of cyclists, which result in sparser and less consistent point cloud representations. As the difficulty increases, environmental occlusion and scene complexity further degrade the detection performance. Nevertheless, both methods maintain a reasonable level of accuracy across all difficulty settings, with SECOND demonstrating a comparatively better performance, particularly in the easy condition.

Fig. 6 illustrates the mAP versus epochs for the Pedestrian class under easy, moderate, and hard detection conditions. From the trends observed in the figure, it is evident that the

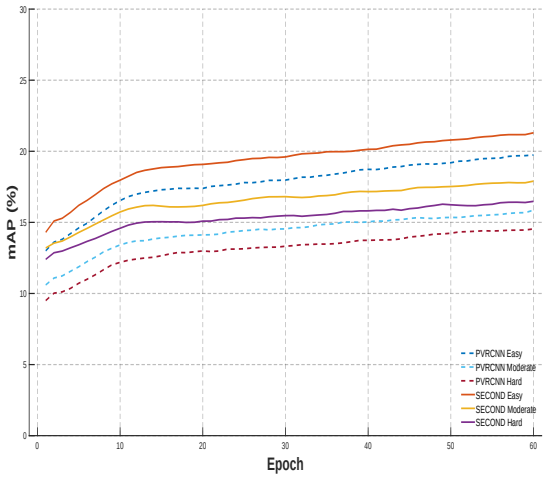


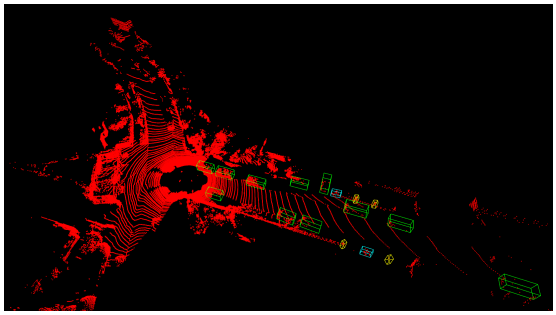
Fig. 6: Pedestrian Class mAP vs Epochs for PVRCNN and SECOND

system's mAP performance steadily improves with increasing epochs and gradually reaches saturation. For the easy condition, PVRCNN achieved a maximum mAP of 19.76%, while SECOND attained a slightly higher value of 21.35%. Under moderate conditions, PVRCNN reached 15.86% and SECOND achieved 17.94%. Similarly, for the hard condition, PVRCNN recorded a mAP of 14.5% and SECOND reached 16.48%. The higher mAP values under easy conditions can be attributed to fewer occlusions and a more distinct point cloud structure, whereas the moderate and hard conditions suffer from increased clutter and reduced point density, which negatively impact detection performance.

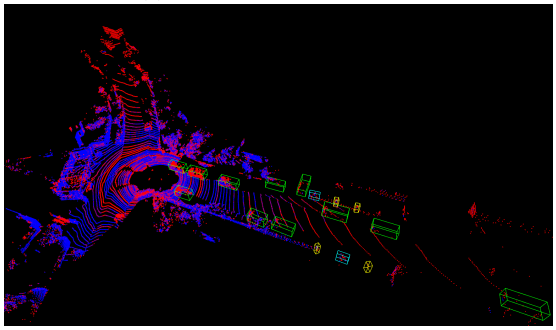
The fig. 7 illustrates point cloud data under two different weather conditions. Subfigure (a) shows the scene under clear weather, where the majority of points are colored red and green, representing structures and bounding box annotations without environmental interference. In contrast, subfigure (b) depicts the same scene under a smoggy condition, which is a combination of fog and aerosol effects. In this case, additional blue and purple points appear in the point cloud, indicating the presence of smog particles (fog and aerosols). These additional points represent environmental noise and disturbances that affect the clarity of bounding box detection and object representation in the point cloud.

V. Concluding Remarks

This work presented a comprehensive study on the joint influence of fog and aerosol particle size distributions on



(a) Clear Weather Condition



(b) Smog(Fog+Aerosol) Condition

Fig. 7: Bounding Boxes Visualization in different environments: (7.a.) Shows Bounding Boxes visualization in Clear weather condition (7.b.) Shows Bounding Boxes visualization in Smog(Fog+Aerosol) condition

LiDAR-based 3D object detection systems. By modeling realistic smog conditions through the combined statistical distributions of fog and aerosol particles, and integrating these effects into the KITTI dataset, we systematically evaluated the degradation patterns in detection performance across varying particle characteristics.

Experimental results consistently demonstrated that the particle size distribution exerts a non-negligible impact on LiDAR returns, directly affecting key detection metrics such as mean Average Precision (mAP) and recall. Specifically, it was observed that broader and more heterogeneous particle distributions impose increased signal attenuation and scattering, thereby challenging conventional detectors not trained under such conditions. Conversely, detectors trained with particle-size-augmented datasets exhibited improved robustness and adaptability to adverse atmospheric conditions.

The findings of this study underscore the critical importance of modeling environmental degradations with high fidelity, particularly in safety-critical applications such as autonomous driving and robotic perception. Beyond dataset augmentation, the results advocate for the development of PSD-aware detection frameworks that can adaptively compensate for varying atmospheric conditions in real time.

References

[1] T. Zhang, X. Huang, and H. Zhao, "Uncertainty-aware 3d object detection for autonomous driving," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 8, pp. 4914–4925, 2021.

[2] M. Chen, B. Xu, J. Wang, and L. Zhang, "Denoising lidar point clouds under adverse weather for robust 3d object detection," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 1, pp. 77–88, 2023.

[3] H. Wang, Y. Liu, and M. Zhang, "Physics-inspired data augmentation for robust lidar perception in adverse weather," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 5, pp. 3451–3462, 2022.

[4] Y. Li, Z. Sun, and L. Zhao, "Adaptive lidar perception under dynamic weather via spatiotemporal consistency," *IEEE Transactions on Intelligent Vehicles*, 2024, early Access.

[5] K. Yang, C.-W. Feng, and X. Zhu, "Multi-task learning for environment-aware 3d object detection using lidar," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 12, pp. 9120–9132, 2022.

[6] P. Sun, H. Kretzschmar, M. d'Arcy, H. He, X. Cheng, B. Wang, Y. Cao, Y. Zhang, M. Chang, Y. Chai *et al.*, "Rsn: Range sparse net for efficient, accurate lidar 3d object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 5725–5734.

[7] X. Bai, X. Chen, and D. Lin, "Pointaugmenting: Cross-modal augmentation for 3d object detection," in *International Conference on Learning Representations*, 2023.

[8] J. Wang, J. Song, W. Jiang, D. Yang, S. Tulsiani *et al.*, "Occflow: Occupancy flow estimation for motion forecasting in dynamic scenes," *arXiv preprint arXiv:2303.11169*, 2023.

[9] M. Bijelic, T. Gruber, F. Mannan, F. Kraus, W. Ritter, and K. Dietmayer, "Seeing through fog without seeing fog: Deep multimodal sensor fusion in unseen adverse weather," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11 682–11 692, 2020.

[10] C. Sakaridis, D. Dai, and L. Van Gool, "Semantic foggy scene understanding with synthetic data," *International Journal of Computer Vision*, vol. 129, no. 5, pp. 1785–1802, 2021.

[11] R. Heinzler, P. Schindler, A. Seekircher, and W. Ritter, "Weather influence and classification with automotive lidar sensors," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 5975–5987, 2022.

[12] K. Kilic, M. Günther, and G. Trommer, "Lidar sensor model for adverse weather conditions," *Remote Sensing*, vol. 13, no. 6, p. 1122, 2021.

[13] M. Hähner, C. Sakaridis, D. Dai, and L. Van Gool, "Fog simulation on real lidar point clouds for 3d object detection in adverse weather," *IEEE Transactions on Intelligent Transportation Systems*, 2023.

[14] X. Peng, R. Li, Y. Zhang *et al.*, "Parallel lidars meet the foggy weather: A multi-sensor approach for robust perception," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1234–1241, 2024.

[15] C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles*. Wiley, 1983.

[16] G. Mie, "Contributions to the optics of turbid media, particularly of colloidal metal solutions," *Annalen der Physik*, vol. 330, no. 3, pp. 377–445, 1908.

[17] V. Wulfmeyer, R. M. Hardesty, D. D. Turner, A. Behrendt, M. P. Cadeddu, P. Di Girolamo, P. Schlüssel, and J. Van Baelen, "Ground-based and airborne lidar remote sensing of atmospheric processes," *Bulletin of the American Meteorological Society*, vol. 99, no. 7, pp. 1303–1330, 2018.

[18] N. Venugopal, S. Zhang, V. Narayanan, P. Boulanger, and S. L. Waslander, "Adverse weather perception using physics-based lidar simulation," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2020, pp. 3022–3031.

[19] S. R. Richter, V. Vineet, S. Roth, and V. Koltun, "Playing for benchmarks," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2213–2222.

[20] M. M. Hnewa and H. Radha, "Object detection in adverse weather using synthetic data augmentation," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 7, pp. 4484–4495, 2021.

[21] S. Liu, J. Li, S. Yu, and L. Xiong, "Simulating adverse weather conditions for autonomous driving," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022, pp. 1234–1243.

[22] T. Elperin, A. Fominykh, and B. Krasovitov, "Effect of raindrop size distribution on scavenging of aerosol particles from gaussian air pollution plumes and puffs in turbulent atmosphere," *Process Safety and Environmental Protection*, vol. 102, 04 2016.

[23] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 7464–7475.

[24] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," *arXiv preprint arXiv:2410.17725*, 2024.